# Fe

## Ridge lease R-07: what the block model contains

Cut-off 45% Fe · existing drilling (14 holes) · model seed 2655

**All data in this report is synthetic.**
The geology, assays, drill holes, prices and costs were generated by a computer model from a fixed random seed to illustrate how a block model works. They do not describe any real place or deposit. This is not a mineral resource or reserve estimate under UNFC, JORC, NI 43-101 or any other reporting code, and must not be presented as the result of field work, sampling or laboratory assays.

## Summary

Above the 45% Fe cut-off the model holds 55.3 million tonnes at an average estimated grade of 59.7% Fe, with a tonnage-weighted mean confidence of 0.54. 60.9% of that tonnage is at G2 or better (confidence of 0.5 or more), and 23.1% is G4, ground with almost no drilling nearby.

| Resource above cut-off | G1 | G2 | G3 | G4 |
|---|---|---|---|---|
| Million tonnes | 19.0 | 14.6 | 8.9 | 12.8 |

## 1. Rock types: how much, how rich, how sure

Share is the percentage of all 408.4 Mt of rock in the model. Grade is the tonnage-weighted estimated iron content. Confidence (0 to 0.97) reflects distance to the nearest drill hole. Chance is the average probability that a cell of that rock beats 45% Fe, given its grade uncertainty.

| Rock type | Mt | Share | Fe grade | Confidence | Chance |
|---|---|---|---|---|---|
| Hematite | 34.5 | 8.4% | 62.9% | 0.52 | 97.6% |
| Blue dust | 6.1 | 1.5% | 64.8% | 0.53 | 98.7% |
| Lateritic ore | 11.7 | 2.9% | 50.9% | 0.65 | 82.5% |
| BHQ (banded hematite quartzite) | 147.8 | 36.2% | 36.8% | 0.50 | 15.5% |
| Waste (phyllite, shale) | 208.4 | 51.0% | 14.0% | 0.19 | – |

Share of each rock type's tonnes by confidence stage:

| Rock type | G1 | G2 | G3 | G4 |
|---|---|---|---|---|
| Hematite | 32.5% | 26.4% | 13.2% | 28.0% |
| Blue dust | 33.6% | 29.6% | 11.7% | 25.1% |
| Lateritic ore | 44.3% | 28.9% | 26.8% | 0.0% |
| BHQ (banded hematite quartzite) | 31.7% | 22.1% | 16.2% | 30.0% |
| Waste (phyllite, shale) | 5.3% | 7.9% | 23.2% | 63.6% |

## 2. Ore above 45% Fe by type of occurrence

| Occurrence | Mt | Of resource | Fe grade | Confidence | G2+ |
|---|---|---|---|---|---|
| Reef ore | 30.9 | 56.0% | 63.0% | 0.52 | 59.6% |
| Pocket deposit | 3.5 | 6.4% | 61.9% | 0.46 | 52.2% |
| Blue dust | 6.1 | 11.1% | 64.8% | 0.53 | 63.2% |
| Lateritic ore | 11.1 | 20.1% | 51.3% | 0.66 | 74.2% |
| Low-grade BHQ | 3.5 | 6.4% | 46.1% | 0.38 | 34.5% |

Pocket deposits are hematite bodies of fewer than 70 connected cells; larger bodies are reef ore. 143 float-ore fragments are mapped on the slopes below ore outcrops.

## 3. Grade and tonnage at other cut-offs

| Cut-off | Mt | Fe grade | Mt at G2 or better |
|---|---|---|---|
| 40% Fe | 90.5 | 52.8% | 51.8 |
| 45% Fe | 55.3 | 59.7% | 33.6 |
| 50% Fe | 47.1 | 61.9% | 29.2 |
| 55% Fe | 42.5 | 62.9% | 25.9 |
| 58% Fe | 39.7 | 63.3% | 24.1 |
| 60% Fe | 34.3 | 64.0% | 20.7 |
| 62% Fe | 25.5 | 65.0% | 15.0 |
| 64% Fe | 16.8 | 66.0% | 9.9 |

## 4. Drill holes

| Hole | Type | East | North | Depth | Best intercept |
|---|---|---|---|---|---|
| DDH-01 | Core | 125 m | 425 m | 170 m | 40 m at 63.4% Fe from 110 m |
| DDH-02 | Core | 250 m | 350 m | 190 m | 40 m at 65.4% Fe from 110 m |
| RC-03 | Reverse circulation | 350 m | 375 m | 160 m | 60 m at 60.7% Fe from 0 m |
| DDH-04 | Core | 475 m | 325 m | 200 m | 30 m at 56.1% Fe from 0 m |
| DDH-05 | Core | 575 m | 375 m | 200 m | None above 45% Fe |
| RC-06 | Reverse circulation | 675 m | 275 m | 200 m | 20 m at 48.6% Fe from 0 m |
| DDH-07 | Core | 800 m | 275 m | 170 m | 20 m at 66.7% Fe from 80 m |
| DDH-08 | Core | 925 m | 250 m | 190 m | 30 m at 62.1% Fe from 70 m |
| RC-09 | Reverse circulation | 200 m | 525 m | 160 m | 30 m at 58.6% Fe from 130 m |
| DDH-10 | Core | 425 m | 450 m | 180 m | 10 m at 53.3% Fe from 10 m |
| DDH-11 | Core | 725 m | 400 m | 160 m | 70 m at 63.9% Fe from 0 m |
| RC-12 | Reverse circulation | 325 m | 200 m | 160 m | None above 45% Fe |
| DDH-13 | Core | 625 m | 175 m | 170 m | None above 45% Fe |
| DDH-14 | Core | 875 m | 125 m | 160 m | 10 m at 45.5% Fe from 30 m |

## How the numbers are made

- Grid of 44 × 32 × 20 cells, each 25 m × 25 m × 10 m. Tonnes = volume × density (waste 2.6, BHQ 3.3, hematite 4.2, blue dust 3.2, laterite 3.0 t/m³).
- Hematite and blue dust are iron oxide (Fe2O3, about 70% Fe when pure). BHQ is hematite banded with quartz, so its iron content is lower. Laterite is a weathered, iron-rich surface cap.
- Confidence decays with horizontal distance to each hole (exp(-d²/3.3²), zero beyond 7 cells) and below the hole's bottom; the top two cells get at least 0.3 from surface mapping.
- Estimated grade departs further from the model's true grade where confidence is low. Uncertainty sigma = 1.5 + 11 × (1 - confidence) % Fe; chance above cut-off = normal CDF of (grade - cut-off) / sigma.
- Stages follow the spirit of the UNFC geological axis: G1 at confidence 0.75 or more, G2 at 0.5 or more, G3 at 0.25 or more, otherwise G4. Figures use existing drilling only.